

```
import pandas as pd
import numpy as np

from google.colab import files

uploaded = files.upload()

filename = list(uploaded.keys())[0]
df = pd.read_csv(filename)

print("Dataset loaded successfully!")
print(df.head())
print(df.columns)
```

movies(1).csv

movies(1).csv(text/csv) - 643 bytes, last modified: 9/19/2026 - 100% done

Saving movies(1).csv to movies(1) (1).csv

Dataset loaded successfully!

	Genre	Director	Actor	Rating
0	Action	Christopher Nolan	Tom Hanks	8.8
1	Drama	Steven Spielberg	Leonardo DiCaprio	8.6
2	Comedy	Rajkumar Hirani	Aamir Khan	8.1
3	Romance	Karan Johar	Ranbir Kapoor	7.5
4	Action	James Cameron	Leonardo DiCaprio	8.7

Index(['Genre', 'Director', 'Actor', 'Rating'], dtype='object')

```
# Remove missing values
df = df.dropna()

# Select features and target
X = df[["Genre", "Director", "Actor"]]
y = df["Rating"]

print("Data preprocessing completed!")
print(X.head())
```

Data preprocessing completed!

	Genre	Director	Actor
0	Action	Christopher Nolan	Tom Hanks
1	Drama	Steven Spielberg	Leonardo DiCaprio
2	Comedy	Rajkumar Hirani	Aamir Khan
3	Romance	Karan Johar	Ranbir Kapoor
4	Action	James Cameron	Leonardo DiCaprio

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
print("Training data:", len(X_train))
print("Testing data:", len(X_test))
```

```
Training data: 12
Testing data: 3
```

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestRegressor

preprocessor = ColumnTransformer([
    ("cat", OneHotEncoder(handle_unknown="ignore"),
     ["Genre", "Director", "Actor"])
])

model = Pipeline([
    ("preprocessor", preprocessor),
    ("regressor", RandomForestRegressor(
        n_estimators=100,
        random_state=42
    ))
])

model.fit(X_train, y_train)

print("Model trained successfully!")
```

```
Model trained successfully!
```

```
from sklearn.metrics import mean_absolute_error, r2_score

predictions = model.predict(X_test)

mae = mean_absolute_error(y_test, predictions)
r2 = r2_score(y_test, predictions)

print("Mean Absolute Error:", mae)
print("R2 Score:", r2)
```

```
Mean Absolute Error: 0.31800000000000005
R2 Score: 0.18372400000000006
```

```
new_movie = pd.DataFrame({
    "Genre": ["Action"],
    "Director": ["Christopher Nolan"],
    "Actor": ["Tom Hanks"]
})
```

```
predicted_rating = model.predict(new_movie)
```

```
print("Predicted Movie Rating:",  
      round(predicted_rating[0], 2))
```

Predicted Movie Rating: 8.37